Mohamed Fathy Mohamed

.NET Tech Lead · Multi-Tenant SaaS Architecture · Microservices · Blazor · Angular

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PROFESSIONAL SUMMARY

Technical Lead and software architect with **9+ years of experience** designing and delivering enterprise web applications on the Microsoft stack, including **3+ years leading the architecture of a multi-tenant ERP SaaS platform** built on .NET 10 microservices. Responsible for the platform's core foundations: tenant lifecycle and data isolation, identity and access management with Keycloak, licensing, event-driven integration and observability. Combines deep hands-on expertise in Clean Architecture, CQRS and Domain-Driven Design with frontend architecture in Blazor WebAssembly and Angular. Leads a development team through technical design, code review and delivery.

9+ years	3+ years	6 gateways	Up to 85%
.NET enterprise development	Leading a multi-tenant ERP	Payment integrations	Less frontend CSS

PROFESSIONAL EXPERIENCE

Technical Lead / Principal .NET Engineer

Jan 2024 – Present

MicroTec — Cloud ERP SaaS · Cairo (serving the Saudi market)

- Lead the architecture of a **multi-tenant ERP platform** built as .NET microservices on **Clean Architecture, CQRS and DDD**; designed the multi-tenancy layer and led the upgrade to **.NET 10 / EF Core 10**.
- Architected **tenant lifecycle management** (version pinning, controlled rollouts, migrations, dedicated-database isolation) and **ESS licensing** across services.
- Own **identity on Keycloak** (custom themes and providers, multi-account sign-in, v23 → v26 migration) and **event-driven integration** with RabbitMQ/MassTransit and the Outbox pattern.
- Established platform **observability** (OpenObserve, Sentry, Elastic) and led frontend architecture across **Blazor WebAssembly** and the **Angular monorepo**.
- Author shared **NuGet packages**; own **Azure DevOps CI/CD** and Azure Container Apps deployments; lead code reviews and mentor engineers.

.NET Developer

Jun 2023 – Dec 2023

MicroTec

- Developed core ERP modules across the .NET microservices backend and Angular frontend; promoted to Technical Lead within seven months.

.NET Web Developer

Jan 2017 – May 2023

ACS — Software House · Cairo

- Designed, developed and maintained **ASP.NET (Core, MVC, WebForms)** applications for government bodies, NGOs and enterprise clients, including GIZ, the Ministry of Health, the Ministry of Education, NCCM and the Federation of Egyptian Industries.
- Delivered secure **payment integrations** with Fawry, QNB, PayPal, E-Finance, Moyasar and Apple Pay for banking, utilities and charity platforms.
- Managed IIS deployments and production support, and published companion iOS and Android applications.

CORE COMPETENCIES

Backend: C# · .NET 10 / ASP.NET Core · EF Core · Clean Architecture · CQRS · DDD · Microservices · Multi-tenancy · RabbitMQ / MassTransit · Outbox · Hangfire · Ocelot · Redis · REST APIs

Identity & Security: Keycloak · OAuth 2.0 / OpenID Connect · JWT · Permission-based authorization

Frontend: Blazor WebAssembly · MudBlazor · .NET MAUI · Angular · PrimeNG · TypeScript · Design systems

Data: SQL Server · PostgreSQL · Oracle (PL/SQL) · MySQL · SQLite

Cloud, DevOps & Observability: Azure DevOps CI/CD · Azure Container Apps · Docker · OpenObserve · Sentry · Elastic

Practices: Testing: xUnit · bUnit · Playwright · Code review · Technical leadership · AI-assisted engineering

KEY PROJECTS & ACHIEVEMENTS

MICROTEC — CLOUD ERP PLATFORM

Multi-Tenant ERP SaaS Platform

Technical Lead · Architect

A cloud ERP serving businesses in Saudi Arabia, built from the ground up as **domain-driven .NET microservices** covering accounting, sales, purchasing, inventory, fixed assets, HR and payroll, and workflow. Designed the multi-tenancy model (tenant resolution, per-tenant data access and caching), the inter-service messaging backbone and the shared package ecosystem that every service builds on, enabling the team to ship new modules on a consistent foundation.

Stack: .NET 10 · EF Core · SQL Server · RabbitMQ/MassTransit · Redis · Ocelot · Hangfire · Keycloak · Docker · Azure Container Apps

Tenant Control Center

Architect · Lead Developer

An operations console that manages the **full lifecycle of every tenant**: per-tenant version pinning with routing of service calls to the pinned version, controlled version rollouts, database migrations, and extraction of a tenant from a shared database into a **dedicated database**. Includes subscription status, audit trail, bulk operations and fine-grained per-action permissions.

Stack: ASP.NET Core · Blazor WebAssembly · EF Core · SQL Server

Employee Self-Service Licensing

Architect · Lead Developer

A licensed Employee Self-Service offering spanning the HR, licensing and identity services. Delivered **seat reservation, invitation, acceptance and revocation** flows, a dedicated identity client so mobile apps hold no secrets, rate-limited activation, and sign-in by email or **international phone number**.

Stack: .NET · Keycloak · MassTransit integration events · Blazor

Identity & Access Modernization

Owner

Led the platform's identity layer on Keycloak: branded bilingual login themes, custom providers for **multi-account sign-in** across tenant realms, a selective **v23 → v26 realm migration toolkit** with dry-run support, and audience-scoped token validation so each API only accepts tokens issued for it.

Stack: Keycloak · OAuth 2.0 / OIDC · PowerShell · Playwright

Central Service Stock & Billing

Architect · Lead Developer

A central ledger for resold third-party services (**OCR, SMS and WhatsApp**) covering purchases, consumption, stock movements and **low-stock alerts**. Designed local-first so reads and purchases never depend on downstream services, with snapshots and consumption synchronized asynchronously through integration events.

Stack: .NET · CQRS · RabbitMQ · Blazor WebAssembly · xUnit

Platform Observability Suite

Architect · Developer

A unified **System Logs** experience backed by configurable providers (OpenObserve, Sentry, Elastic), with service-dependency graphs, a per-tenant activity timeline, token-claim inspection and client-side error capture, reducing the time needed to trace a failed request across services.

Stack: OpenObserve · Sentry · Elastic · ASP.NET Core · Blazor

HR & Payroll on Blazor WebAssembly

Technical Lead

Led the migration of the HR module from Angular to **Blazor WebAssembly**: payroll runs, payments and actions, loans, allowances and employee reports, with **permission-based authorization**, saved table views, keep-alive tabs, org chart, calendar, dark mode and push notifications, all on a shared design system.

Stack: Blazor WebAssembly · MudBlazor · .NET · bUnit

Frontend Design System

Technical Lead

Introduced a Figma-aligned **design-token layer and dark mode** across the Angular monorepo, consolidating duplicated styling into shared components and cutting stylesheet code by **43–85% per application**. Added a Ctrl/Cmd+K command palette with Arabic-aware search, and a matching Blazor component library.

Stack: Angular · PrimeNG · SCSS design tokens · Blazor component library

Workflow Designer

Technical Lead

A visual designer for business **approval workflows** (steps, actors, actions and effects), sold as an optional ERP module. Rebuilt the interface on the shared ERP design system with a step-timeline editor, feature-based structure and component tests.

Stack: Blazor WebAssembly · ASP.NET Core · bUnit

EARLIER CLIENT PROJECTS — ACS

ESMC Prepaid Meter Recharge
Developer

A recharge portal that reads customer meter cards and processes top-ups, integrated with the **Fawry, BEE and E-Finance** payment networks.

EBI Payment Platform
Developer

The end-to-end payment process for the **Egyptian Banking Institute**, integrated with QNB.

DSCA Donations Platform (KSA)
Developer

A public donations platform for a Saudi charity, integrated with **Moyasar and Apple Pay**.

GIZ Education & Case Management
Developer

Ministry of Education dashboards on vocational education across Egypt, and a beneficiary case-management system with ministry-level reporting.

NCCM Child Protection
Developer

A public website and iOS/Android apps for reporting missing children and child abuse, connected to a police-facing portal.

Wessam — Ministry of Health
Developer

A clinic-quality rating platform with three-level surveys, yearly progress tracking and awards.

Also delivered: Petro Gulf Medical System · EFE Grants Management · Federation of Egyptian Industries portal · Egyptian Swimming Federation rankings.

INDEPENDENT PROJECTS

Mirsad
Creator · Live product

An AI assistant that answers business questions in **Arabic or English** from any SQL database or REST API; every generated query passes a **guard, permission check and tenant scoping**.

Stack: .NET 10 · Microsoft.Extensions.AI · React · Angular & Blazor widgets · SignalR · Docker

Dawsha
Creator · Open source

Runs **Claude Code, Codex and Gemini** side by side in readable terminals on Windows, macOS and Linux, with agent timelines and **remote control from a phone**.

Stack: Electron · TypeScript · xterm.js · node-pty · Vite · Playwright

Nadfny
Creator · Open source

A Windows disk cleaner for developers: **git worktrees, Docker, build output and caches**, with guards that never delete unsaved work or source code.

Stack: .NET 10 · WebView2 · WiX · xUnit

Notes Tracker
Creator · Live product

A cross-platform notes app for **Windows, Android and web** with offline sync, voice notes with speech-to-text transcription, text extraction from images, shared notebooks and a full Arabic right-to-left interface.

Stack: .NET 10 · .NET MAUI · Blazor · ASP.NET Core · PostgreSQL · SQLite

AI Engineering Toolkit
Creator

A set of AI agent skills that diagnose production issues: container health on Azure, **tracing a failed request by correlation ID**, auditing Keycloak realms and inspecting SQL databases, used for day-to-day production support.

Stack: Claude Code · Azure CLI · OpenObserve · Playwright

Learning Hub
Creator

Self-contained engineering lessons, browsable as an animated, searchable **mind map**.

Stack: HTML · CSS · JavaScript · GitHub Pages

EDUCATION

B.Sc. Computer and Information Science
Ain Shams University, 2016
Graduation project (Very Good): audio source separation of multi-instrument recordings.
IBM Summer Boot Camp — hybrid mobile apps, IBM MobileFirst (2015).

LANGUAGES

Arabic — Native
English — Fluent (professional working proficiency)